Lab 3 Data Preparation

CRISP-DM Methodology

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Use this as an alternative to a phone-less assessment for the data preparation section and fill out the new cases

[1] Reviews on Flipkart

1.1 Dataset

1.2 Initial EDA (explore and collect errors) on Flipkart data

```
In [1]: import pandas as pd
        df = pd.read_csv('flipkart_data.csv')
```

```
In [2]: df.shape
Out[2]: (205052, 10)
```

```
In [3]: df.info()
<class 'RangeIndex'>
RangeIndex: 205052 entries, 0 to 205051
Data columns (total 10):
 product_name  205052 non-null  object
 product_price 205052 non-null  object
 rate          205052 non-null  object
 review        205052 non-null  object
 summary       205052 non-null  object
 sentiment     205052 non-null  int64
 sentiment.1   205052 non-null  int64
 sentiment.2   205052 non-null  int64
 sentiment.3   205052 non-null  int64
 sentiment.4   205052 non-null  int64
dtype: int64, object: 6
memory usage: 15.6 MB
```

```
In [4]: df.head(2)
Out[4]:
  product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
```

```
In [5]: df.tail(2)
Out[5]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
```

1.3 Missing values

```
In [6]: df.isnull().sum()
Out[6]: product_name    0
        product_price   0
        rate            0
        review          0
        summary         0
        sentiment       0
        sentiment.1     0
        sentiment.2     0
        sentiment.3     0
        sentiment.4     0
        dtype: int64
```

```
In [7]: df.duplicated().sum()
Out[7]: 0
```

1.4 Unique values

```
In [8]: df.nunique()
Out[8]: product_name    1
        product_price   1
        rate            1
        review          1
        summary         1
        sentiment       1
        sentiment.1     1
        sentiment.2     1
        sentiment.3     1
        sentiment.4     1
        dtype: int64
```

1.5 Value counts

```
In [9]: df['product_name'].value_counts
Out[9]: <bound method IndexOpsMixin.value_counts of 0    Candes Baby
        1    Candes Baby
        2    Candes Baby
        3    Candes Baby
        4    Candes Baby
        ...
        205046  Candes Baby
        205047  Candes Baby
        205048  Candes Baby
        205049  Candes Baby
        205050  Candes Baby
        205051  Candes Baby
        Name: product_name, Length: 205052, dtype: object>
```

```
In [10]: df['product_price'].value_counts
Out[10]: <bound method IndexOpsMixin.value_counts of 0    Rs.1,000.00
         1    Rs.1,000.00
         2    Rs.1,000.00
         3    Rs.1,000.00
         4    Rs.1,000.00
         ...
         205046  Rs.1,000.00
         205047  Rs.1,000.00
         205048  Rs.1,000.00
         205049  Rs.1,000.00
         205050  Rs.1,000.00
         205051  Rs.1,000.00
         Name: product_price, Length: 205052, dtype: object>
```

```
In [11]: df.rate.value_counts
Out[11]: <bound method IndexOpsMixin.value_counts of 0    5
         1    5
         2    5
         3    5
         4    5
         ...
         205046  5
         205047  5
         205048  5
         205049  5
         205050  5
         205051  5
         Name: rate, Length: 205052, dtype: object>
```

```
In [12]: df.review.value_counts
Out[12]: <bound method IndexOpsMixin.value_counts of 0    Very
         1    Very
         2    Very
         3    Very
         4    Very
         ...
         205046  Very
         205047  Very
         205048  Very
         205049  Very
         205050  Very
         205051  Very
         Name: review, Length: 205052, dtype: object>
```

```
In [13]: df.summary.value_counts
Out[13]: <bound method IndexOpsMixin.value_counts of 0    Very
         1    Very
         2    Very
         3    Very
         4    Very
         ...
         205046  Very
         205047  Very
         205048  Very
         205049  Very
         205050  Very
         205051  Very
         Name: summary, Length: 205052, dtype: object>
```

```
In [14]: df.sentiment.value_counts
Out[14]: 1    205052
         Name: sentiment, dtype: int64
```

```
In [15]: df.sentiment.value_counts
Out[15]: 1    205052
         Name: sentiment, dtype: int64
```

```
In [16]: df.sentiment.value_counts
Out[16]: 1    205052
         Name: sentiment, dtype: int64
```

```
In [17]: df.sentiment.value_counts
Out[17]: 1    205052
         Name: sentiment, dtype: int64
```

```
In [18]: df.sentiment.value_counts
Out[18]: 1    205052
         Name: sentiment, dtype: int64
```

1.6 Dropping

1.7 Summarizing the data based on column

```
In [19]: df.describe(include='object')
Out[19]:
       product_name product_price  rate  review  summary
count        205052        205052  205052  205052  205052
unique            1             1       1       1       1
top     Candes Baby  Rs.1,000.00       5    Very    Very
freq         205052        205052  205052  205052  205052
```

```
In [20]: df.describe(include='int64')
Out[20]:
       sentiment  sentiment.1  sentiment.2  sentiment.3  sentiment.4
count   205052.0     205052.0     205052.0     205052.0     205052.0
mean         1.0          1.0          1.0          1.0          1.0
std          0.0          0.0          0.0          0.0          0.0
min          1.0          1.0          1.0          1.0          1.0
25%          1.0          1.0          1.0          1.0          1.0
50%          1.0          1.0          1.0          1.0          1.0
75%          1.0          1.0          1.0          1.0          1.0
max          1.0          1.0          1.0          1.0          1.0
```

1.8 Dropping columns

```
In [21]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[21]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

1.9 Renaming columns

```
In [22]: df.rename(columns={'sentiment': 'rate'})
Out[22]:
       product_name product_price rate review summary rate
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
...             ...          ...  ...    ...    ...   ...  ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
[205052 rows x 6 columns]
```

1.10 Replacing

```
In [23]: df.replace('Very', 'Very Very')
Out[23]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
...             ...          ...  ...    ...    ...   ...       ...       ...       ...       ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
[205052 rows x 10 columns]
```

1.11 Replacing

```
In [24]: df.replace('Very', 'Very Very')
Out[24]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
...             ...          ...  ...    ...    ...   ...       ...       ...       ...       ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
[205052 rows x 10 columns]
```

1.12 Replacing

```
In [25]: df.replace('Very', 'Very Very')
Out[25]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
...             ...          ...  ...    ...    ...   ...       ...       ...       ...       ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
[205052 rows x 10 columns]
```

[2] Data Cleaning and Preparation

Urvil, Jinjina, Rhea

2.1 Dataset

```
In [26]: import numpy as np, pandas as pd, matplotlib.pyplot as plt, seaborn as sns
         df = pd.read_csv('flipkart_data.csv')
```

2.2 Dimensions

```
In [27]: df.shape
Out[27]: (205052, 10)
```

2.3 Columns

```
In [28]: df.columns
Out[28]: Index(['product_name', 'product_price', 'rate', 'review', 'summary',
                'sentiment', 'sentiment.1', 'sentiment.2', 'sentiment.3',
                'sentiment.4'],
                dtype='object')
```

2.4 Dropping columns

```
In [29]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[29]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.5 Renaming columns

```
In [30]: df.rename(columns={'sentiment': 'rate'})
Out[30]:
       product_name product_price rate review summary rate
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
...             ...          ...  ...    ...    ...   ...  ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very    1
[205052 rows x 6 columns]
```

2.6 Replacing

```
In [31]: df.replace('Very', 'Very Very')
Out[31]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
...             ...          ...  ...    ...    ...   ...       ...       ...       ...       ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
[205052 rows x 10 columns]
```

2.7 Replacing

```
In [32]: df.replace('Very', 'Very Very')
Out[32]:
       product_name product_price rate review summary sentiment sentiment.1 sentiment.2 sentiment.3 sentiment.4
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
...             ...          ...  ...    ...    ...   ...       ...       ...       ...       ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1         1         1         1         1
[205052 rows x 10 columns]
```

2.8 Dropping

```
In [33]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[33]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.9 Dropping

```
In [34]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[34]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.10 Dropping

```
In [35]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[35]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.11 Dropping

```
In [36]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[36]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.12 Dropping

```
In [37]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[37]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.13 Dropping

```
In [38]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[38]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.14 Dropping

```
In [39]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[39]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.15 Dropping

```
In [40]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[40]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.16 Dropping

```
In [41]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[41]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.17 Dropping

```
In [42]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[42]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.18 Dropping

```
In [43]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[43]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.19 Dropping

```
In [44]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[44]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.20 Dropping

```
In [45]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[45]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.21 Dropping

```
In [46]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[46]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.22 Dropping

```
In [47]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[47]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.23 Dropping

```
In [48]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[48]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.24 Dropping

```
In [49]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[49]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
4       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
...             ...          ...  ...    ...    ...   ...       ...
205046  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205047  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205048  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205049  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205050  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
205051  Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
[205052 rows x 6 columns]
```

2.25 Dropping

```
In [50]: df.drop(['sentiment.1', 'sentiment.2', 'sentiment.3', 'sentiment.4'], axis=1)
Out[50]:
       product_name product_price rate review summary sentiment
0       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
1       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
2       Candes Baby  Rs.1,000.00   5   Very  Very  Very         1
3       Cand
```